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SARDAR PATEL UNIVERSITY

M. Sc (Information Technology) Semester-I Examination (NC)

PS01CINT54 : Relational Database Management Systems

Friday, 12th April 2024

Time: 10.00 am to 1.00 pm

Total Marks:70

Q-1 Select an appropriate option for each of the following statement:

[08]

- 1 Which of the following is not the type of the data model?
[a] Relational Data Model [b] Semi-structured Data Model
[c] Entity-Relationship Model [d] Process-Based Model
- 2 Which of the following is not a type of database?
[a] Hierarchical [b] Network [c] Distributed [d] Decentralized
- 3 Weak entity set is represented as _____.
[a] Underline [b] Double Underlined [c] Double Rectangle [d] Double Diamond
- [a] Relationship [b] Association [c] Set [d] Combination
- 4 _____ is an example of one-to-many relationships considering Teacher and Class relationship.
[a] One class may have many teachers [b] One teacher can have many classes
[c] Many classes may have many teachers [d] Many teachers may have many classes
- 5 _____ command is used to change values of records in a table.
[a] ALTER TABLE [b] INSERT [c] UPDATE [d] None of given
- 6 _____ is an output of MOD(9,2)
[a] 2 [b] 1 [c] 0 [d] None of given
- 7 The _____ clause is used in conjunction with the GROUP BY clause to impose a condition on it.
[a] WHERE [b] ORDER BY [c] HAVING [d] None of given
- 8 Execution of _____ is done automatically
[a] Function [b] Procedure [c] Trigger [d] a and c both

Q-2 Answer the following questions in brief. (Attempt Any SEVEN)

[14]

- 1 What is DBMS? Write application areas of Database.
- 2 Explain concept of Data Independency.
- 3 How RDBMS is differed from DBMS?
- 4 Explain briefly at least FOUR notations used to draw E-R diagram.
- 5 Differentiate clearly stored and derived attributes.
- 6 Explain with example the built in function floor().
- 7 Write SQL command to add primary key ENO in an existing table EMP.
- 8 Explain briefly with example(s) the table DUAL.
- 9 State TWO major differences between FUNCTION and Stored PROCEDURE.

- Q-3 Do as directed.**
[A] Categorized users of DBMS in detail. [06]
[B] Write a note on Data Models. [06]

OR

- [B] Discuss need, advantages and disadvantages of DBMS in detail. [06]
Q-4 Do as directed.
[A] Explain the concept of normalization. Discuss 1NF and 2NF with appropriate example. [06]
[B] List out Codd's rules and explain any three of them. [06]

OR

- [B] What is key? Discuss different types of keys in DBMS. [06]

Q-5 Do as directed.

- [A] Write SQL command for each of the following. [Any FOUR] [06]
Assume that the table EMP has fields (ENO number(2,0), DNO number(1,0), ENAME varchar2(15), BPAY Number(8,0))
i. To create a table EMP having primary key ENO.
ii. To display unique records of a table EMP.
iii. To display DNO and DNO wise total BPAY for all employees of a EMP table.
iv. To delete all the records from a table EMP having DNO greater than 3.
v. Retrieve all employee information of table EMP in descending order of ENAME.
[B] Explain briefly following statement with example of your choice: [06]
a. FOR statement b. UPDATE Statement

OR

- [B] Explain briefly following built-in functions with example(s) of your choice: [06]
a. SUBSTR() b. ROUND()

Q-6 Do as directed.

- [A] Write short note on PL/SQL block structure. [06]
[B] Write short note on Cursor, including its definition, types and attributes. [06]

OR

- [B] Assume that table named TELE has fields TELENO Number(5), CNAME varchar2 (20), NCALLS number(5,0) and BMT number (10,0). [06]
Write a PL/SQL code block to read a TELENO from the user, if given TELENO found in a table TELE then display the message 'RECORD FOUND', otherwise display the message that 'RECORD NOT FOUND'.